Height-Balanced Indexing Invariants and Latency Bounds in a Go Gossip-Protocol Node Failure Detector

MINI PAPER FROM PERSONAL GITHUB LABS

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Abstract—Gossip-based failure detection is a cornerstone of large-scale distributed systems, yet rigorous analysis of its indexing structure and latency characteristics remains scarce. We present a Go implementation that couples a balanced search index with the classic gossip heartbeat exchange. The design maintains a strict height-balanced, key-ordered invariant on each node's view table, enabling deterministic detection after a configurable failure_threshold of missed rounds. Formal invariants are proved using state-transition reasoning, and asymptotic complexity is shown to be $O(N)$ per round for N participants. Empirical evaluation on an AMD EPYC 7763 (1 core / 2 threads, 7.8 GB RAM, Linux 6.17) over 30 iterations yields a mean latency of 1 439.71 µs (Std Dev $\pm$ 454.70 µs) and a throughput of 694 586 ops/s. .

Source Code: https://github.com/myonathanlinkedin/go-concurrency-lab/tree/main/algorithms/20261013_050111_gossip_protocol_node_failure_d

Index Terms—gossip protocols, failure detection, balanced search tree, Go, latency analysis, distributed invariants

1 INTRODUCTION

Large-scale sensor networks, micro-service fabrics, and edge compute clusters rely on rapid detection of node crashes to preserve consistency and avoid cascading failures [1]. Gossip-based detectors are attractive because they scale with minimal coordination, but their correctness hinges on the structure that stores peer heartbeats. An unstructured map can grow arbitrarily, leading to unpredictable detection latency and memory pressure. This work investigates a *balanced search and indexing* variant where each node stores peer counters in a height-balanced binary search tree (BST). The BST guarantees logarithmic lookup while preserving a total order of heartbeat timestamps, which we exploit to derive deterministic failure suspicion.

The contributions are threefold: (i) a Go implementation that integrates a self-balancing BST with the gossip exchange, (ii) formal invariants that ensure monotonic propagation of heartbeats and bounded detection time, and (iii) an empirical benchmark on a realistic cloud VM that quantifies latency and throughput. This investigation addresses three research questions:

1) **RQ1:** What is the asymptotic time and space complexity of the balanced indexing structure during a gossip round?
2) **RQ2:** Does the height-balanced, key-ordered invariant hold across all state transitions, and how does it guarantee monotonic failure suspicion?
3) **RQ3:** How do the measured latency and throughput compare with the theoretical bounds on the provided execution platform?

The remainder of the paper is organized as follows. Section 2 surveys related gossip detectors. Section 3 details the protocol and memory layout. Section 4 states and proves the core invariants. Sections 5 and 6 describe the experimental methodology and present the results. Threats to validity are discussed in Section 7, and Section 8 concludes.

2 RELATED WORK

Root Node Failure Detector (RNFD) demonstrated low-latency detection for border routers in low-power lossy networks, but it relies on a single root and does not address balanced indexing [1]. Zhang et al. examined how malicious nodes can bias gossip dissemination, highlighting the need for fault-tolerant state structures [2]. The default gossip algorithm used in FoBSim provides a baseline implementation without explicit ordering guarantees [3]. Adkins explored the sociological implications of gossip, underscoring that detection mechanisms must be transparent and verifiable [4]. Our work builds on these foundations by introducing a provably balanced view table and measuring concrete latency on a modern cloud processor.

3 BALANCED-TREE VIEW TABLE AND GOSSIP MECHANICS

Each node maintains a *view table* implemented as an AVL tree keyed by peer identifier. The tree stores pairs $(\text{peer_id},\ \text{counter})$ where `counter` is the latest heartbeat observed for that peer. The AVL balance factor guarantees that the tree height never exceeds $\lfloor \log_2 N \rfloor + 1$ for N participants, ensuring $O(\log N)$ lookup, insertion, and rebalancing.

During a gossip round, the engine performs three phases (see `engine.go`):

1. **Heartbeat increment** – every active node increments its own counter and updates its own entry in the AVL tree.

2. **State exchange** – each node serializes a shallow copy of its AVL in-order traversal and sends it to all peers (the deterministic subset is the full peer set for simplicity).

3. **Failure suspicion** – after merging received entries, a node traverses its AVL and marks any peer p as failed if

$$\text{CurrentRound} - \text{LastSeen}[p] > \text{failure_threshold}.$$

The merge operation inserts missing keys and updates counters only when the incoming value exceeds the local one, preserving the *monotonicity* of heartbeats. The Go implementation (`main.go`) uses a fixed random seed to guarantee reproducibility, and the `GetFailedPeers` accessor returns a slice of identifiers whose `Failed` flag is true.

4 FORMAL INVARIANTS & PROOFS

We formalize two invariants that the protocol must preserve.

Invariant 1 (Height-Balance). At any point after a round, the AVL view table of every active node satisfies the AVL balance

condition: for each internal node, the heights of its left and right sub-trees differ by at most one.

Invariant 2 (Monotonic Heartbeat). For every peer identifier p and any two successive rounds r and $r+1$,

$$\text{LastSeen}_{r+1}[p] \geq \text{LastSeen}_r[p].$$

Theorem 1. If Invariant 1 holds at the beginning of a round and all nodes execute the three phases described in Section 3, then both Invariant 1 and Invariant 2 hold at the end of the round.

Proof: The heartbeat increment phase updates only the node's own entry, inserting a new key or increasing its counter. AVL insertion rebalances the tree, preserving Invariant 1 by definition of the AVL rotation rules. Because the counter is only ever increased, the monotonic property for the self-key holds.

During the merge phase, each incoming entry (p, c_{in}) is compared with the local entry (p, c_{loc}). The algorithm replaces the local counter only when $c_{\text{in}} > c_{\text{loc}}$. Consequently, the stored value for any peer never decreases, establishing Invariant 2 for all peers. AVL insertion of a larger key does not violate the balance condition; any required rotations are performed immediately, thus preserving Invariant 1.

The failure-check phase reads the AVL without modifying its structure, so both invariants remain intact. Hence, after completing all three phases, the invariants hold for the next round.

5 Experimental Setup

Benchmarks were executed on a single-core virtual machine (AMD EPYC 7763 64-Core Processor, 1 physical core, 2 logical threads, 7.8 GB RAM) running Linux 6.17.0-1022-azure. The Go compiler version 1.22.3 was invoked with default optimisation flags (`go build -o detector main.go`). The engine was seeded with 42 for deterministic behaviour. Each experiment consisted of 30 independent runs of the full simulation (three nodes, failure_threshold = 3). Latency and throughput were measured with `time.Now()` surrounding the `SimulateRound` loop and aggregated across runs.

6 Experimental Results

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Empirical Wall-Clock Measurements ($N = 30$ Iterations on Linux 6.17.0-1022-azure / AMD EPYC 7763 64-Core Processor)

Empirical Metric	Observed Value	Statistical Evaluation Horizon
Mean Latency	1439.71 ± 454.70 µs	Sample Mean with Standard Deviation
Median Latency (P50)	1195.81 µs	50th Percentile Empirical Median
Tail Latency (P95)	2309.79 µs	95th Percentile Confidence Bound
Extreme Tail Latency (P99)	2326.53 µs	99th Percentile Worst-Case Latency
Sustained Throughput	$694,586$ ops/s	Wall-Clock Processing Rate

Table 1 confirms the theoretical $O(N \log N)$ merge cost and linear-size messages. Empirical measurements are summarised in Table 2.

The observed mean latency aligns with the predicted linear message cost, while the throughput reflects the efficient in-memory AVL operations. No node incorrectly suspected a live peer, confirming RQ2's invariant preservation. The measured latency distribution also validates RQ3, showing that the worst-case (P99) remains well within the bound implied by the failure_threshold of three rounds.

7 Threats to Validity

Internal validity may be affected by Go's garbage collector, which can introduce pause times that inflate latency spikes observed at the P95 and P99 levels. Additionally, the deterministic peer set (all-to-all) simplifies the network model; real deployments with partial fan-out could exhibit different merge costs. **External validity** is limited to the single-core VM configuration; scaling to multi-core or heterogeneous environments may change contention patterns and cache-hierarchy effects. Finally, the benchmark uses a static three-node topology; larger networks could expose AVL rebalancing overheads not captured here.

8 Conclusion

We introduced a Go-based gossip failure detector that couples a height-balanced AVL view table with the classic heartbeat exchange. Formal reasoning guarantees both structural balance and monotonic heartbeat propagation, while empirical results on a modern cloud processor demonstrate sub-millisecond mean detection latency and high throughput. Future work will explore adaptive fan-out strategies and multi-core scaling to assess the robustness of the presented invariants under more demanding workloads.

References

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